

Horner's Method for Evaluating Polynomials

All iterative methods require the evaluation of functions for which we are trying to find the solutions. Since it is a recurring task, the design of an efficient algorithm for evaluating the function is important. While it is not possible to propose a general algorithm for evaluating transcendental equations, it is quite simple to design an algorithm for evaluating polynomials.

Suppose a polynomial equation is expressed as,

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$$

where a_k are real numbers representing the polynomial coefficients and x^k are the powers of the polynomial variable x . The above polynomial is said to be of **degree n** , where n represents the highest variable exponent.

If we try to evaluate the polynomial equations as it is, it will take $\frac{n(n+1)}{2}$ multiplications and n addition operations. So, the total number of operations will be:

$$\begin{aligned} & \frac{n(n+1)}{2} + n \\ &= \frac{n^2 + n}{2} + n \\ &= \frac{n^2}{2} + \frac{n}{2} + n \\ &= O(n^2) \end{aligned}$$

[You do not need to write the following proof in examination; this is added just for your own visualization purpose]

Why $\frac{n(n+1)}{2}$ number of multiplications and n number of additions? We can figure it out from an example,

Let's consider a polynomial equation, $f(x) = 2X^3 - 4X^2 + 10X - 7 \dots\dots$ (i)

$$\underbrace{2X^3 - 4X^2 + 10X}_{\text{3 additions}} - 7 = 0$$

It is easy to see from this equation that there are 3 additions $\rightarrow$ one less than the number of terms. Also, 3 is the degree of this equation. So, we can say we need n number of addition operations.

As for the multiplication,

Term	No. of required multiplication
X	0 (1 - 1)
X ²	X * X = 1 (2 - 1)
X ³	X * X * X = 2 (3 - 1)
....	
X ⁿ	n - 1

So for, $X + X^2 + X^3 + \dots + X^n$, no. of required multiplication,

$$\begin{aligned} &0 + 1 + 2 + \dots + (n - 1) \\ &= \frac{(n - 1)(n - 1 + 1)}{2} \\ &= \frac{(n - 1)(n)}{2} \end{aligned}$$

We have yet to multiply the terms with their coefficients, so for n degree equation, it will take n more multiplications. So, total number of required multiplications:

$$n + \frac{(n - 1)(n)}{2} = \frac{n(n + 1)}{2}$$

Why use Horner's method?

From the above proof, we can see the reason behind higher multiplication costs is the increasing power of the variable x . Each term requires multiple multiplications to compute the power of x . If we can somehow rearrange the equation in a form, so that x is factored incrementally, then each term involves only a single multiplication with x . This way, we only need to handle the cost of multiplying with the coefficients.

Let's look at the multiplication and addition cost after rearranging equation (i),

$$\begin{aligned} & 2X^3 - 4X^2 + 10X - 7 \\ &= X(2X^2 - 4X + 10) - 7 \\ &= X(X(2X - 4) + 10) - 7 \end{aligned}$$


Here, we need **exactly 3 multiplications** – which is the number of degree of this equation. As for the number of additions.

$$X(X(2X - 4) + 10) - 7$$


We need a total of **3 (degree of the eqn)** additions here too. So, the total number of required multiplication and addition operations will be:

$n + n = O(n)$ [Time complexity is improved]

General form of Horner's Method,

$$f(x) = (((.....((a_n x + a_{n-1})x + a_{n-2})x + + a_1)x + a_0$$

Recursive Rule:

$$p_n = a_n$$

$$p_{n-1} = p_n * x + a_{n-1}$$

.....

.....

$$p_1 = p_2 * x + a_1$$

$$f(x) = p_0 = p_1 * x + a_0$$

[Note: This process starts by assigning the highest-degree coefficient a_n to p_n . Then, at each step, we multiply the current result by x and add the next lower coefficient. This continues until we reach p_0 , which gives us the final value of the polynomial $f(x)$. This way, each step involves exactly one multiplication and one addition.]

Let's try to evaluate the polynomial $f(x) = 2x^3 - 4x^2 + 10x - 7$ at $x = 1$ using Horner's Method

$$p_3 = a_3 = 2$$

$$p_2 = p_3 * x + a_2 = 2 * (1) - 4 = -2$$

$$p_1 = p_2 * x + a_1 = -2 * (1) + 10 = 8$$

$$p_0 = p_1 * x + a_0 = 8 * (1) - 7 = 1$$

So, for $f(1)$, the result is **1**, and we can see that Horner's method also gives the same value.

Algorithm:

Step 1: Take the *degree* of the polynomial as input, **n**

Step 2: Input the value of x to evaluate the function with

Step 3: Initialize the coefficients $a_0, a_1, a_2, \dots, a_n$

Step 4: Initialize a variable **result** with a_n

Step 5: for ($i = n - 1; i \geq 0; i--$) {

$$\text{result} = \text{result} * x + a_i$$

}

Step 6: Print Result

Example 1: Evaluate the polynomial $f(x) = x^3 - 4x^2 + x + 6$ using Horner's rule at $x = 2$

Here, $n = 3, a_3 = 1, a_2 = -4, a_1 = 1, a_0 = 6$

$$p_3 = a_3 = 1$$

$$p_2 = 1 * (2) + (-4) = -2$$

$$p_1 = (-2) * 2 + 1 = -3$$

$$p_0 = (-3) * 2 + 6 = 0$$

So, $f(2) = 0$

Self-study: Evaluate the polynomial $f(x) = x^4 + 2x^2 - 16x + 5$ using Horner's rule at $x = 2$.

References:

- "Numerical Methods" by E. Balagurusamy, Publisher: Tata McGraw-Hill, 4th Ed., 1999.